A Comprehensive Survey of Adaptive Strategies in Differential Evolutionary Algorithms

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Abstract

Classical differential evolution (DE) encounters premature convergence when dealing with diverse optimization problems. This challenge has encouraged extensive research efforts aimed at improving and enhancing the original methodologies. Among the various improvement techniques, adaptive strategies have been universally employed. However, there is a lack of systematic research on the adaptation mechanisms. This work comprehensively investigates the adaptive strategies adopted in DE algorithms. Typical adaptation strategies employed in DE algorithms are refined and summarized, highlighting their characteristics. A new taxonomy of adaptation strategies is proposed, categorizing them based on their primary properties, which include adaptations of control parameters, mutation strategies, population size, search space, learning schemes, and composite adaptations. The advantages and disadvantages of these adaptation strategies are summarized, elucidating their unique characteristics. Additionally, a general framework with an adaptive updating engine is proposed, which can serve as a reference for developing new DE algorithms or improving existing ones. The paper also highlights the challenges and open issues of adaptive strategies, suggesting several promising research directions.

Keywords: Adaptive strategy, adaptive mechanism, adaptation strategy, evolutionary algorithm, differential evolution.

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1. Introduction

Although differential evolution (DE) [1] was proposed three decades ago, research on it keeps hot and ongoing. DE was put forward by Storn and Price [1] in 1995, which is both concise and powerful in dealing with global numerical problems. DE and its variants have constructed significant branches of evolutionary computation (EC), which employs mutation, crossover and selection operations. However, classical DE has its limitations. The scaling factor F and the crossover rate CR are manually selected and fixed throughout the entire evolution. This constraint can hinder the algorithm performance as different parameter values may be needed during exploration and exploitation. Furthermore, trial-and-error selection of parameters F and CR is time-consuming. In the past two decades, the limitations of the classical DE algorithm have encouraged researchers to search for better solutions. Some researchers have reported that DE variants can be improved by tuning the parameters during evolution. However, the trial-and-error approach to select appropriate parameters requires tedious trials and plenty of time. If the parameters can be adaptively adjusted according to the characteristics of the problem, the algorithm will be more likely to demonstrate robust performance [2, 3, 4].

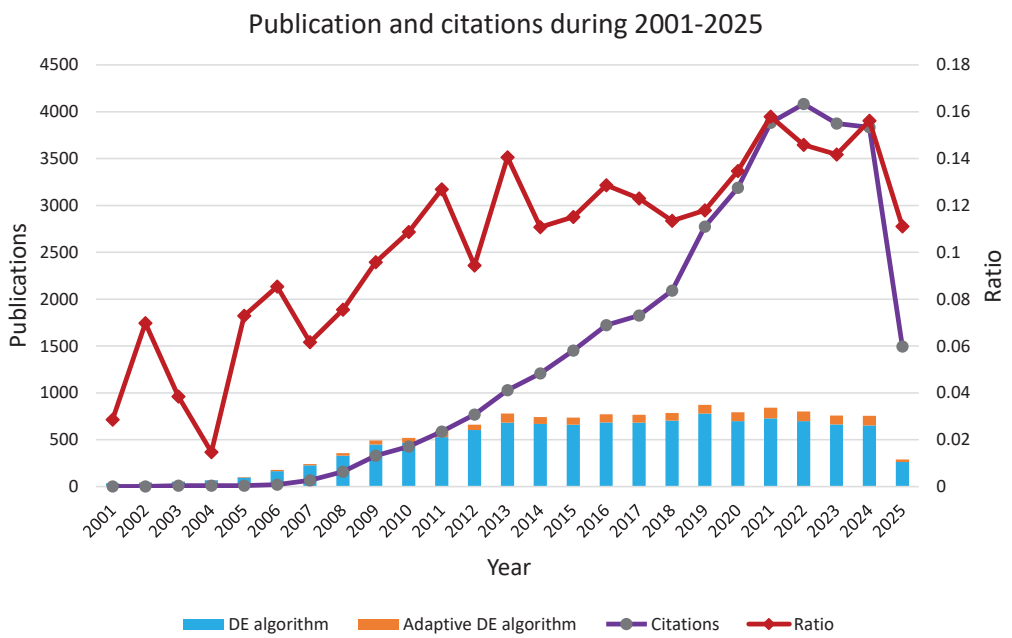

Fig. 1. Number of publications, citations and ratio (adaptive DE algorithms/DE algorithms) from 2001 to 2025, the data are from the Web of Science.

As illustrated in Fig. 1, the statistical data from the Web of Science reveal a significant

upward trend of DE and adaptive DE algorithms in the publications and citations over the past two decades. Although there was a slight decrease in publications and citations in 2022 and 2023, the ratio of adaptive DE publications relative to total DE publications rises quickly from 2023 to 2024. As more adaptive DE algorithms are proposed, there is a growing need to categorize them, identify their characteristics, and reveal the implication of adaptation strategies.

Based on extensively surveying hundreds of papers on adaptive DE algorithms, the most frequently employed adaptation strategies can be further categorized according to their main characteristics: adaptive control parameters (F and CR, we suggest not including population size NP due to its independent features.) [2, 3, 4, 5], adaptive mutation scheme (frequently including target vector and donor vector) [6, 7, 8, 9], adaptive population size [4, 10, 11, 12], adaptive search space (landscape between the lower and upper bounds) [13, 14, 15, 16], adaptive learning scheme [17, 18, 19, 20], and composite adaptation (using more than three types of adaptive strategies mentioned above.) [4, 7, 8, 10]. The category graph can be shown in Fig. 2.

There are several survey papers on the development of DE and its variants. Das and Suganthan [21] conducted a comprehensive study of various state-of-the-art DE algorithms, analyzing the parameters, mutation schemes, complex optimization, drawbacks, and real-world engineering applications of DE algorithms. Later, Das *et al.* [22] proposed a survey of DE algorithms, summarizing and organizing several developments such as adaptive parameters in DE variants, DE algorithms for single objective problems, applications in various optimization scenarios, and hybridization of DE and other algorithms. Suganthan *et al*. [23] comprehensively reviewed DE variants based on adaptation techniques, surveyed six adaptive DE variants for single objective optimization, and outlined several applications of adaptive DE variants, such as constrained optimization, multi-objective optimization, and large-scale optimization. Karafotias *et al.* [24] investigated the development of parameter control strategies in EAs, including the parameter setting, selection, controlling and ensembles of various techniques. Al-Dabbagh *et al*. [25] reviewed 28 adaptive DE algorithms and classified them into two categories: adaptive parameter with single DE mutation and adaptive parameter with multiple DE mutations. Meanwhile, the authors categorized the DE strategies into standard and advanced DE strategies. Wang *et al.* [26] surveyed the employment of different parameter configurations, considering F, CR, NP and M, and proposed a self-adaptive ensemble-based DE approach. Recently, Tanabe and Fukunaga [27] investigated 24 DE variants based on parameter control methods (PCMs), in which the population size was regarded as an independent parameter to be researched.

This paper is distinguished from the survey papers mentioned above. The differences and contributions are as follows (see Table1).

Table 1
The differences among our paper and other review papers.

Ref.	Main content	Year
[21]	1) Survey of several state-of-the-art DE algorithms, and their applications to different complex problems; 2) Analyze the DE variants based on control parameters, mutation schemes; 3) Theoretical study of DE, including its drawbacks; 4) Offer potential directions of DE; 5) Overview of real-world applications with DE algorithms.	2011
[23]	1) Additional trial vector generation; 2) Analyze recent adaptive DE variants, such as jDE, SaDE, JaDE and etc; 3) Adaptations for multi-objective, large-scale, constraint, dynamic problems; 4) Adaptive population topology.	2014
[24]	1) Review the parameter control mechanisms in EAs; 2) Discuss methods for controlling multiple parameters; 3) Overview the trends of the field.	2015
[22]	1)survey the development of DE algorithms; 2) Parameter adaptation in DE algorithms; 3) Single-objective optimization based on DE algorithms; 4) DE for different fields applications; 5) Hybridization of DE and other methodologies.	2016
[25]	1) Extend the taxonomy of parameter settings in EAs; 2) Provide a metaheuristic framework with parameter control; 3) Propose two taxonomies of parameters in DE algorithms; 4) Establish a procedural analysis over 28 adaptive DE algorithms; 5) Reveal the pros and cons of different adaptive scheme; 6) Provide a framework to devise a metaheuristic algorithms based on parameter control.	2018
[26]	1) Investigate the impact of different adaptive parameter settings by evaluating on 34 benchmark functions; 2) Devise an algorithm based on control parameters adaptation and ensemble strategies (SAEDE); 3) Reveal the influence of the combination of different parameters, such as NP, F, CR, and mutation scheme.	2020
[27]	1) Review 24 parameter control methods (PCMs) in DE variants, providing a systematic category; 2) PCMs focus on F and CR; 3) Conduct an extensive benchmarking study of PCMs; 4) Discuss the relationships among PCMs in DE; 5) Analyze the performance of PCMs by comparing with recent algorithms;	2020
Our	1) Survey of DE algorithms based on adaptive strategies; 2) Provide a new taxonomy of six adaptive strategies; 3) Research the main operations of adaptive strategies in DE variants; 4) Reveal the main impacts of adaptive strategies; 5) Highlight the advantages and disadvantages of adaptive strategies; 6) Provide a framework of devising a DE algorithm based on six categories of adaptive strategies; 7) Analyse the challenges, open issues, and promising directions of categorized adaptive strategies.	2025

The remainder of this paper is organized as follows. Section 2 provides a brief introduction to the classical DE, followed by demonstrations of adaptive adjustments of control parameters in Section 3, mutation strategies in Section 4, population size in Section

5, search space in Section 6, learning schemes in Section 7, and the composite strategy in Section 8. Section 9 compares and summarizes the categorized adaptive strategies. Challenges, open issues, and promising directions for adaptive strategies are highlighted in Section 10, suggesting a general framework to devise a new or improve an existing adaptive (DE) algorithm. Section 11 concludes this work.

2. The overview of differential evolution

The population in DE algorithm includes NP individuals $P = \{\mathbf{x}_1^g, \mathbf{x}_2^g, ..., \mathbf{x}_{NP}^g\}$ in the gth generation. Without losing generality, the optimization problems in this paper are to minimize the objective problem $f(\mathbf{x})$, which is defined by:

$$Minimize\ f(\mathbf{x}),\ s.t.\ \mathbf{x} = (x_1, x_2, ..., x_D) \in \mathbb{R}^{\mathbb{D}} \tag{1}$$

where $x_i \in [l_i, u_i],\ i = 1, ..., D$, l_i and u_i are the minimum and maximum values of the search domain, respectively.

2.1. Mutation

Generally, the DE strategy can be written as the form: $DE/a/b/c$ [28], in which a refers to the target vector, b denotes the difference vector number, and c is the type of crossover such as binomial and exponential. DE employs a mutation strategy to generate a trial vector $\mathbf{v}_i$ associated with the target vector $\mathbf{x}_i$. The most frequently adopted mutation strategy include DE/rand/k (A1) [28, 29, 30, 31], DE/best/k (A2) [19, 32, 33, 30], DE/current-to-best/k (A3) [31, 34, 35, 36], DE/current-to-rand/k (A4) [5, 12, 30, 37, 38], DE/rand-to-best/k (A5) [19, 39, 40] and etc.

2.2. Crossover

The crossover operation is used to enhance the diversity of the population [28]. According to the crossover probability CR, a new trial vector u_i can be obtained from a binomial or exponential crossover. The former can be described as:

$$u_{i,j}^{g+1} = \begin{cases} v_{i,j}^g & r_j \leq \text{CR} \mid\mid j = j_r \\ x_{i,j}^g & otherwise \end{cases} \tag{2}$$

where $r_j \in$ (0,1), $CR \in (0, 1)$, j_r is a randomly selected number within the set $\{$1, 2, ..., $NP\}$. The CR controls a small portion of information inherited from the mutant vector.

The latter can be described as [41]:

$$u_{i,j}^{g+1} = \begin{cases} v_{i,j}^g & \forall j = \{\langle t \rangle_D, \langle t+1 \rangle_D, ..., \langle t+L-1 \rangle_D\} \\ x_{i,j}^g & otherwise \end{cases} \tag{3}$$

where $\langle \cdot \rangle_D$ is to modulo D, t is a crossover operation starting point, $t \in$ [1,D], L is the consecutive length of crossover operation that $L \leq D$ and t+L-1 should be less than D.

2.3. Selection

The selection strategy follows a greedy process. The trial vector $\mathbf{u}_i^g$ competes with the parent vector $\mathbf{x}_i^g$ according to the fitness value $f(\cdot)$, and the winner will remain as the parent in the next generation of evolution. The selection operation can be described as follows.

$$\mathbf{x}_i^{g+1} = \begin{cases} \mathbf{u}_i^g & if\ f(\mathbf{u}_i^g) \leq f(\mathbf{x}_i^g) \\ \mathbf{x}_i^g & otherwise \end{cases} \tag{4}$$

The three operations mentioned above are generally kept fixed and repeatedly conducted to generate a new population until the termination criteria are satisfied.

2.4. An overview of adaptive strategies in DE algorithms

There are many survey papers [21, 22, 23, 24, 25, 26] tracking and reporting the state-of-the-art development of DE. Many techniques have been adopted to improve or design new DE variants. Adaptive strategies are frequently employed in these improvement methodologies. However, the adaptive strategies are rarely categorized, and their characteristics including advantages, disadvantages, and potential directions have not been fully researched. To fill these gaps, this work focuses on the adaptation mechanisms employed in DE variants.

Based on extensive survey on adaptive DE algorithms, the adaptation strategies can be categorized into six classes: control parameter adaptation, adaptive mutation strategy, population size adaptation, search space adaptation, adaptive learning strategy, and composite adaptations. In order to intuitively illustrate the taxonomy of the adaptive strategies, a graphical framework is suggested in Fig. 2. It illustrates that each category of adaptive strategies incorporates many techniques. The applications and characteristics of each category will be comprehensively analyzed in the following sections.

3. Control parameter adaptation

The control parameters in this paper include F and CR, because they directly impact the information in offsprings. The population size will be analyzed independently. Methods for adaptively updating F and CR can be manifold.

3.1. Stochastic approaches

Qin *et al.* [5] reported that the CR was sensitive to the multimodality of the problem, while F was much more likely to impact the convergence speed. The authors developed a self-adaptive DE algorithm (SaDE) in which the parameters F and CR were dynamically generated by normal distributions with fixed parameters i.e. F=$N(0.5, 0, 3)$, and CR was adaptively yielded by CR=$N(\mu_{CR,k}, 0.1)$, where $\mu_{CR,k}$ is the median value of

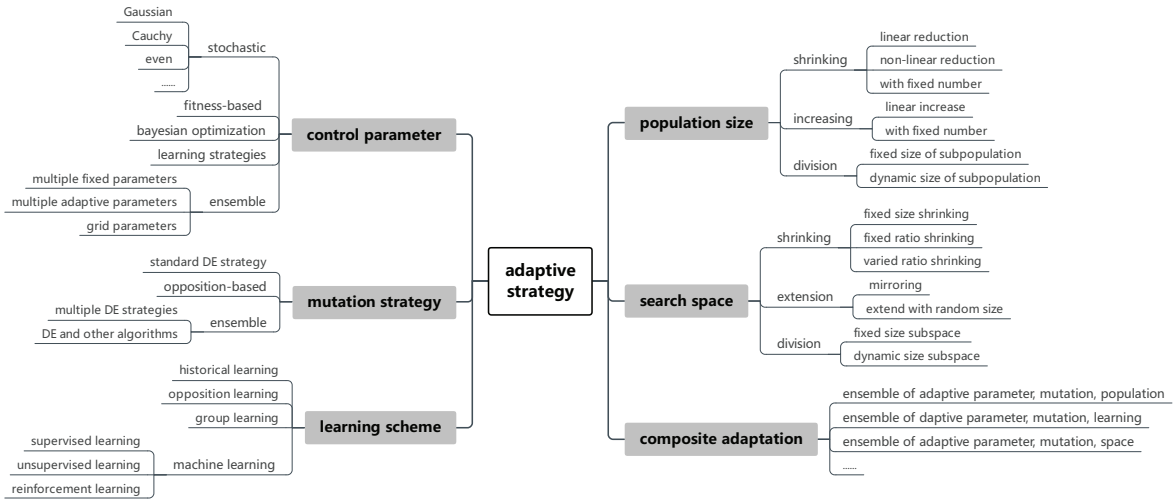

Fig. 2. A taxonomy of adaptive strategies employed in DE algorithms.

the CR which obtains better offsprings within consecutive N_c generations. However, the experimental results indicated that parameter adaptation could not significantly improve the function evaluation, especially on multimodal functions [5]. Instead of utilizing fixed parameters in the normal distribution of F in [5], Zhang and Sanderson [3] updated the scaling factor in JADE by Cauchy distribution, in which the expected value μ_F was initially defined 0.5 and updated by (A10), CR is adaptively adjusted by truncated normal distribution as in (A12). Cauchy mutation helps diversify the mutation operations and prevent premature convergence.

To improve the robustness of JADE, Tanabe *et al.* [42] developed Success-history based Adaptive DE (SHADE), which introduced a new parameter adaptation mechanism to record successful parameters had been employed. The mean values of S_F in (A10) and S_{CR} in (A13) in each generation will be maintained in the historical memory of M_F and M_{CR}. The μ_F in (A10) and μ_{CR} in (A12) are updated by M_F (A14) and M_{CR} (A15), respectively. As SHADE provides a diverse set of parameters to guide the evolution, it will not be negatively impacted even if there are poor values of μ_F and μ_{CR} which helps the algorithm to be more robust than JADE.

In order to overcome the bias that the arithmetic mean value in (A13) will make μ_{CR} converge to a small value, Peng *et al.* [43] improved it by a weighted mean as in (A17). Similarly, Choi *et al.* [44] proposed an adaptive Cauchy DE algorithm (ACDE), which encapsulated F and CR in new population generation processes. The values of F and CR were adjusted by a Cauchy distribution based on the average values of F and CR obtained from the successful individuals in the gth generation. Ma *et al.* [45] introduced an adaptive

strategy to diversify the controlling factors, in which F and CR were dynamically changed by normal distributions, see (A19). The empirical results indicated that the randomization processes helped to enrich the diversity of the population, and benefited for the search efficiency.

Unlike many JADE-based algorithms that the standard deviation in the Cauchy and Normal distributions are fixed, the parameters in [46] are adapted according to the fitness values.

$$\begin{cases} F_i = randn_i(M_{F,r_i}, \delta_F) \\ CR_i = randn_i(M_{CR,r_i}, \delta_C R) \end{cases} \tag{5}$$

where δ_{CR} and δ_F are adaptively updated by (A32) and (A33), respectively.

In order to diversify the population, the parameters F and CR in [47] are adaptively adjusted according to the status of each individual as in (A20) and (A21). The empirical results indicated that adaptive parameters helped improve the mutation, crossover and selection, and hence enhance the robustness of the algorithm.

Brest *et al.* [2] thought that better parameters generated superior individuals. In jDE [2], the values of F and CR are adaptively updated during the iteration processes by (A22) and (A23). The values of F and CR with better optimization results will be maintained and propagated into the next generations. Later, Brest *et al.* [48] proposed an improved version of $dynNP - DE$ [49] named $jDEdynNP - F$ to deal with high-dimensional problems, in which the sign of F was implemented to judge the states of improvement. Viktorin *et al.* [50] have found that changing the parameters by distance-based weighted factors (A35) can reward the exploration ability, especially in high-dimensional problem evaluations. Gong [51] *et al.* introduced a repairing crossover rate method, in which a binary string was used to update CR by (A36). Experiments showed that this method enhanced the convergence speed and optimization results. In addition to the linear adjustments mentioned above, there are some techniques proposed for adaptively tuning the parameters. In [52], the control parameters are adaptively changed by two power functions as in (A24) and (A25). The operations can help obtain a large F and a small CR to enhance the population diversity at the beginning of evolution.

3.2. Fitness-based parameter adjustment

The control parameters can also be adjusted by changing the fitness values. Ali and Törn [53] utilized a fitness-based adaptation mechanism to adaptively update the value of F by (A38). To this end, the value of F will be large at the early stages and small at later stages. Similar to jDE [2], F and CR in [54] are updated by comparing the fitness value of the current population with the mean fitness in each generation as in (A39) and (A40). If the fitness is improved, the new parameters will be maintained to propagate new

offsprings. The empirical results indicated that the algorithm with parameter adaptation strategies obtained better solutions than the original method. Ghosh *et al.* [32] found that the neighborhood search of the individuals near the global minimum would be reduced to conduct a better search. They adaptively tuned F and CR based on the fitness values by (A41) and (A43), respectively.

The parameters in [55] are based on both the parameter strategy in JADE and an individual improvement mechanism, which can be seen in (A54) and (A52). The empirical results reveal the robustness and efficiency of the algorithm, which can be enhanced by adopting the proposed parameter adaptation strategies. Tang *et al.* [56] argued that the control parameters should be adjusted to the fitness value of the individual. They proposed two schemes to renew the control parameters. One is the rank-based scheme (A49), in which the individuals are sorted in ascending fitness order. Another scheme is based on the fitness (A50), in which the values of F and CR can be adaptively updated. Meanwhile, in order to improve the algorithm to be more suitable for dealing with multimodal problems, F and CR were modified by normal distributions in [56]. Considering that a particle with better fitness has better optimization ability, Zhou *et al.* [57] assigned a smaller CR to individuals with better fitness value, and vise versa. Experimental results indicate that the sorted CR mechanism help improve the global optimization ability.

Similarly, Cheng *et al*. [58] proposed a cosine distribution scheme to adaptively adjust the scaling factor as in (A47). The scaling factor in (A47) is a nonlinear cosine-based function that when the fitness is poor, F will be enlarged in the next generation, and vice versa. The experimental results indicated that SADE-FP obtained better solutions and converged faster than other techniques. Instead of using successful historical archive, Stanovov and Semenkin [59] proposed the L-SRTDE, the winner of CEC 2024 competition, in which the adaptation of the scaling factor was relied on the success rate. The scale parameter F in L-SRTDE is updated by (A56).

3.3. Heuristic approach

Some researchers have implemented evolutionary approaches to update the control parameters. Salman *et al.*[60] employed the DE/rand/1 strategy to update F, which was empirically proven to be effective in improving the optimization ability of the algorithm. Stanovov *et al.* [61] employed genetic programming (GP) to automatically design the parameters F and CR, and empirically proved that the GP adaptation was effective and beneficial for the algorithm.

In addition, F and CR can be adaptively tuned by fuzzy logic. Lampinen [62, 63] proposed a fuzzy logic controller to adaptively adjust F and CR. The adaptation included two steps: 1) it utilized fuzzy rules to generate a fuzzy output OU, and 2) it employed a defuzzification strategy to generate a real output value. The empirical results indicated faster

convergence of the algorithm. However, FADE was largely impacted by the predefined fuzzy logic sets, which encountered limitations.

Recently, Bayesian Optimization (BO) was validated to be successful in optimizing the parameters of EAs. The mathematical form of BO can be written as follows.

$$P(X|Y) = P(Y|X)P(X)/P(Y) \tag{6}$$

where $P(X|Y)$ is the occurrence probability of event X given the event Y, and vice versa. In [64], BO is employed to successfully control the hyperparameters in multiple adaptive DE (MadDE). In [65], BO is successfully adopted to dynamic parameter calibration and enhances the exploration of the BO-based DE algorithm. The adoption of BO helps the algorithm select proper hyperparameters for solving the Storm Water Management Model (SWMM) problem.

Bui *et al.* [17] found that the variation of F could significantly influence the neighborhood search for the current individuals. A large value of F at the beginning of iteration helps better exploration, while gradually decreasing F will benefit for better exploitation. The decreasing mechanisms can be linear [56], sin/cosine [66], and sigmoidal [17] functions. In Ref. [17], the scaling factor F is updated by (A51).

3.4. Ensemble of control parameter adaptations

Adaptively selecting F and CR from a predefined parameter set is also beneficial for improving an EA. Brest *et al.* [67] employed an ensemble of three pairs of F and CR settings to adaptively generate new offsprings in jDElsgo algorithm (A58). Later, Brest *et al.* [6] improved jDElsgo by implementing three DE strategies in jDEscop: 1) jDEbin (self-adaptive DE/rand/1/bin); 2) jDEexp (self-adaptive DE/rand/1/exp); and 3) jDEbest (self-adaptive DE/best/1/bin). The three DE strategies are encoded into individuals, and each strategy employs an independent set of parameters F and CR as defined in (A59) and (A59), respectively. Based on jDElsgo [67] and jDEscop [6] strategies, Brest *et al.* [7] proposed a new DE strategy called jDEsps, in which three pairs of F and CR settings according to some trigger conditions such as in (A88). The empirical results indicated that the jDEsps algorithm obtained better results on the evaluation of CEC2012 large-scale optimization problems.

Later, Brest *et al*. [68] developed a new algorithm named jSO, in which an ensemble of F settings was predefined to adjust the weight of the trial vector $(x^g_{pbest} - x^g_i)$ as in (A66) to be (A61). The ensemble of F_w values offers a smaller weight for the trial vector in the early stage of evaluation, and a higher weight in the latter stages. The experimental results indicated the superiority of the jSO compared with the former DE variants.

In order to adaptively adjust the search directions, the scaling factor in LSHADE-EpSin [66] is adaptively tuned by a two-stage mechanism. In the first half of the whole

generation, F will be adjusted by sinusoidal decreasing and sinusoidal increasing mechanisms, which can be seen in (A62) and (A63), respectively. In the second half of the generation, F and CR are adjusted by Cauchy and Normal distributions, respectively. The ensemble approach helps the algorithm enhance the convergence speed as well as the global search ability. The idea of GADE (Greedy Adjustment Differential Evolution) [69] is that within a learning period, the current F and CR are compared with their neighbors. F will be randomly selected from the set $\{(F-d_1), F, (F+d_1)\}$, and CR will be chosen from $\{(CR_m - d_2), CR_m, (CR_m + d_2)\}$. F and CR_m are updated by (A44).

Inspired by the local binary pattern (LBP) operator from image processing, Zhao *et al.* [70] proposed a new multimodal optimizer, in which F was adaptively adjusted by the LBP-based niche performance and uniformity of the niche as in (A64), and CR was defined as in (A65). The empirical results indicated that the proposed approach not only enhanced the population diversity but also accelerated the convergence. An adaptive parameter mechanism based on neighborhoods is reported in [71], in which F is declining based on (A94). Similarly, CR is adjusted by (A95) which is based on the distance between X_i and its neighborhood individuals. The empirical results show that the adaptations of F and CR help enhance the search ability of the proposed algorithm.

Apart from the ensemble of adaptations mentioned above, Tatsis *et al.* [72] proposed a grid-based parameter adaptation scheme to update F and CR with the form $Y = X + k * C$ (Y and X can be F or CR, C is the step size, and k is a constant factor). However, the grid-based parameter adaptation significantly increases the computational time and run times. The adaptive control parameter strategies can be summarized in Table A1.

4. Adaptive mutation strategy

Adaptive mutation schemes can be categorized into two main categories, single mutation schemes and ensembles of different mutation strategies.

4.1. Single mutation strategy

The DE standard mutation strategy (A1)-(A5) can be adaptively updated by changing the target vector or donor vector. To improve the performance of DE/rand/1, a new mutation strategy, in which an auxiliary vector amv was added in [73]. The vector amv can be seen in (A80). The experimental data indicated that the auxiliary vectors helped prove the exploration and exploitation (E-E) of the algorithm. A new adaptive mutation scheme based on DE/rand/1 was proposed by A. W. Mohamed and A. K. Mohamed [74], in which the trial vector was changed according to the sorted fitness values. The new strategy can be seen in (A75), which utilizes three vectors from the top $NP \cdot 100p\%$, the bottom $NP \cdot 100p\%$, and the remaining (1-2$\cdot 100p\%)NP$ of the population, respectively. The empirical results indicated that the new strategy helped balance the E-E and enhanced the robustness.

To balance the E-E during the evolution, Bajer [75] proposed an adaptive k-tournament mutation scheme to adaptively change the selection pressure of choosing the base vector $\mathbf{x}_{r1}^{g}$ in (A1). Each individual had its own tournament size k_j which was adaptively changed in the search process. If the trial vector successfully entered the next generation, the tournament size would be updated by (A78). The adaptation was proven to be valuable for reducing the computation and helped to control the E-E abilities.

A new mutation strategy named "DE/current-to-$p$best/1" based on the mutation DE/current-to-best/1 was proposed in JADE [3] as the form (A66). Moreover, the authors utilized an external archive $\mathbf{A}$ to store the inferior solutions, and proposed a new mutation strategy, DE/current-to-$p$best/1 with archive (A67). The empirical results indicated that the implementation of the DE/current-to-$p$best/1 with and without an archive helps to improve the robustness of the algorithm.

Inspired by the DE/current-to-$p$best in JADE [3], a new mutation strategy was proposed by Ramos *et al.* [11], in which a $p$best individual was employed as the base vector, the donor vector was generated by the difference between two individuals randomly selected from the current population and the archive population. The mutant can be seen in (A68). The experimental results indicated that the adaptation strategy benefited the algorithm with better performance on hybrid problems.

Unlike DE/current-to-best/1 which selects the optimal individual to adjust the target vector, Islam *et al.* [76] suggested a new DE strategy named $DE/current\text{-}to\text{-}gr_best/1$ (MDE$_p$BX) which employed the best individual in a randomly combined group ($100q\%$ of the population size) to perturb the parent (target) vector in the current generation, see (A70). Meanwhile, a new biased parent selection mechanism was incorporated by making each mutant cross with one of the $NP \cdot 100p\%$ top-ranked individuals in the current population. With the p-best crossover, the mutant vector is possible to exchange its components with the individual from the top $NP \cdot 100p\%$ population. The experimental results indicated improvements in computational accuracy and effectiveness.

Inspired by the opposition-based learning (OBL) [77], a new mutation strategy based on DE/current-to-$p$best/1 was proposed in [78] which replaced the individual x_{pbest} with the opposite best individual $\mathbf{x}_{opbest}$. The new mutation strategy is defined in (A71). The experimental results indicated improvements in the local search performance.

Recently, many adaptive DE algorithms based on L-SHADE have been developed. Hamdi *et al.* [79] proposed an improved version of L-SHADE which implemented a linearly decreased weight to adapt the target vector in the DE/current-to-$p$best/1, see (A76). The proposed algorithm was successfully implemented to solve the economic environmental power dispatch problems. A new adaptive penalty strategy in L-SHADE was proposed in Ref. [80] , in which the penalty factor (PF) was adaptively adjusted according to the balance of the fitness value $f(x)$ and the constraint violation $v(x)$ as follows.

$$PF = \frac{f(x_i) - f(x_j)}{v(x_i) - v(x_j)} \tag{7}$$

where $i \neq j$, $i, j \in$ 1,...,NP.

To avoid falling into local optima, Zhang and Meng [81] proposed a deeply informed mutation strategy to adjust the search direction as in (A79). The experimental results indicated the method helped accelerate the convergence and prevent stagnation.

4.2. Ensemble of adaptive mutation strategies

It has been found that the "DE/rand/k" scheme has strong exploration ability [82], the "DE/best/k" strategy obtains a high convergence speed [83], the "DE/current-to-$p$best/k" scheme achieves more diversity and greater exploration ability than that of "DE/best/k" [3]. Instead of solely employing "DE/rand/1" in the classical DE, SaDE [5] utilizes two DE strategies, "DE/rand/1" and "DE/current-to-best/2". The authors proposed an adaptive learning strategy to update the selection probability when one of the suitable mutation strategies was selected in the ensemble mutations. The probabilities to select the two candidate strategies are both initially set to 0.5 and then updated by p_1 and p_2 ($p_2 = 1 - p_1$), respectively.

$$p_1 = \frac{ns1 \cdot (ns2 + nf2)}{ns1 \cdot (ns2 + nf2) + ns2 \cdot (ns1 + nf1)} \tag{8}$$

where $ns1$ and $ns2$ are the successive generations that help the parents enter the next generation by employing "DE/rand/1" and "DE/current-to-best/2", respectively. Otherwise, the failure numbers $nf1$ and $nf2$ will be recorded correspondingly. By adaptively learning from p_1 and p_2, the SaDE is able to prevent possible side effects in the previous learning processes.

Similarly, Brest *et al.* [6] proposed an ensemble of mutations which included three mutation schemes, and each mutation strategy would be selected according to the current state "s".

$$\begin{cases} s = 1, & if\ r_1 < 0.1\ \&\ it > \frac{G_{max}}{2} \\ s = 2, & if\ i < 0.5NP \\ s = 3, & otherwise \end{cases} \tag{9}$$

where s=1 denotes the $jDEbest$ strategy, s=2 presents the $jDEbin$ strategy, and s=3 corresponds to the $jDEexp$ strategy, where i=it mod NP. The experimental results provided evidence that the proposed algorithm was highly competitive for large-scale continuous problems. A similar ensemble of mutations was proposed in AVSDE [14] which included three different mutation strategies (A85). The experimental results indicated the better performance of the algorithm with the mutation pool strategy, especially on low dimensional benchmark problems.

Choi *et al.* [84] extended the PSRMM [85] by introducing four DE strategies (4PRMM): 1) DE/rand/1; 2) DE/rand/2; 3) DE/current-to-best/1; and 4) DE/current-to-best/2. The four DE strategies were employed in the order: 2)-1)-3)-4), and were equally allocated for every 1/4 of the maximum evaluations. Meanwhile, the scaling factor was updated when different strategies were employed. In AMECoDE [86], the trial vector is generated by two multiple-elite-guided-based mutation strategies: 1) DE/current-to-nbest/1 (A90); and 2) DE/current-to-$p$best/1. Similarly, three mutations with different characteristics are employed in [38] by collecting 1) DE/current-to-rand/1; 2) DE/rand/1; and 3) DE/current-to-$p$best/1 into an ensemble of mutations. The mutation scheme will be selected by sorting the fitness values of the population. In [87], a donor vector is generated by selecting a mutation strategy from an ensemble of six mutation schemes. In addition, some researchers have attempted to employ some learning strategies to adaptively adjust the selection of proper mutations e.g., a random forest was utilized to adaptively select a proper candidate mutation operator for different optimization problems in [88].

To enhance the convergence performance, an adaptive mutation strategy based on DE and PSO mechanisms (DEPSO) was developed in [89]. The DE/rand/1 mutation and PSO mutation strategies in DEPSO are adaptively selected according to the selection probability SP_G. If *rand*$<SP_G$, the DE/rand/1 mutation mechanism will be selected. Otherwise, the PSO mutation will be employed. The SP_G is defined as

$$SP_G = \frac{1}{1 + e^{1-(G_{max}/g+1)^{\tau}}} \tag{10}$$

where τ is a positive constant. According to SP_G, the *DE/rand/1* and PSO strategies are adaptively applied in the early and later stages of the evolution. In this case, DEPSO can effectively obtain stronger global exploration ability by employing a new strategy based on DE/rand/1, and higher convergence ability by using the PSO mutation strategy. As a consequence, the two strategies help the algorithm enhance the diversity of the population in early evolution, and obtain faster convergence speed in the later stages of the evolution. Similarly, Li *et al*. [90] modified CoDE and JADE to alternatively utilize them to balance the exploration and exploitation.

In order to enhance the E-E of the algorithm, Mostafa *et al.* [91] introduced two hybrid adaptive mutation strategies, combining DE mutation and the spider wasp optimizer (SWO). The two mutations are 1) DE/rand-global-WSO/2 (A91); and 2) DE/rand-local-WSO/2 (A93). The experimental data revealed that the two mutations benefited the algorithm for enhancing the search ability and convergence speed.

Instead of directly selecting a standard DE strategy from the ensemble of mutation strategies, Ref. [92] collects eight standard DE mutations to generate 10 ensemble strategies (ENSs) to form a set of mutations. The ENSs are adaptively selected according to the

historical behavior of the algorithm. The experimental results indicated that the adaptive strategies were beneficial to the large-scale feature selection.

A new approach proposed by Bu *et al.* [55] adopts an ensemble of two mutation strategies. One is employed in the intra-cluster evolution, randomly selecting from four strategies (DE/current-to-best/1, DE/rand-to-best/1, DE/current-to-rand/1, and DE/rand/1). The other is utilized in extra-cluster evolution, which implements DE/current-to-$p$best/1 and DE/rand-to-$p$best/1. The empirical results indicated that the mutation strategies help the algorithm diversify the population and jump out from local optima. In [93], ensemble of three DE strategies are proposed to diversify the population, including 1) DE/current-to-$p$best/1; 2) DE/current-to-crand/1 (A96); and 3) DE/scaled-best/1 (A99). The selection of the three strategies is based on the probabilities each mechanism generates successful offsprings.

Recently, the arm bandit approach has been adopted in EAs to select proper parameters or strategies to generate better offsprings. Huang *et al.* [94] employed the multi-arm bandit (MAB) technique to select different mutation strategies and CRs in each generation. The empirical results indicated that it was beneficial for improving the search ability of the algorithm. In [95], MAB strategy is adopted to adaptively generate new particles in promising areas, which experimentally proved to help reduce the generation number. A K-armed bandit model was developed by Zhang *et al.* [96] to adaptively reduce the computational budget and improve the accuracy. In [97], MAB scheme is adopted to adaptively pick proper direction information for the specific mutation strategy. A probability sw proposed in [98] is employed to adaptively switch the two types of CRs, uniform and binomial to generate proper trial vectors. The parameter sw is generated by fitness rate rank MAB strategy. The adaptive mutation strategies can be summarized in Table A2.

5. Population size adaptation

The population size can be changed by reduction, augmentation and division during the problem evaluations.

5.1. Population reduction

Based on jDE [2], a new DE variant was proposed in [49] with the population size NP decreasing by half in each iteration cycle. In every generation, the individuals from the first half (x_i^g) and the second half ($x^g_{\frac{NP}{2}+i}$) are competing to survive as new parents in the next generation. The population size NP^g is defined by (A100). The experimental results indicated that the dynamic population size reduction scheme could help improve the efficiency and robustness of the algorithm.

An adaptive population tuning scheme (APTS) developed by Zhu *et al.* [99] was successfully adopted to improve DE algorithm, in which the population size (NP) was

adaptively adjusted by an inferior-based population-cut scheme that when the algorithm stagnated NP would be dynamically decreased. If the fitness was repeatedly improved for consecutive K_1 generations, the inferior individuals from the top $NP{\cdot}100p\%$ population would be removed. Meanwhile, if NP equaled to upper bound (UB) for consecutive K_2 generations, the population-cut strategy would be triggered.

Tanabe and Fukunaga [4] improved the SHADE [42] by adopting a linear population size reduction mechanism to diversify the population (L-SHADE). The linear population reduction mechanism is defined as in (A101). Similarly, to keep the population diverse enough, the worst individuals will be periodically deleted by (A117). Meanwhile, the authors employed an archive to preserve $aSize$ best individuals. The population size will be updated by removing inferior individuals periodically.

However, Awad *et al.*[100] suggested that the population reduction in [4] and [66] had two limitations: 1) reduction of the population at the first generation wasted time for initialization and deleting them immediately; and 2) population reduction increased the chance of premature convergence. Hence in [100], a niching-based population reduction strategy was proposed, in which the diversity of the population was measured by the Euclidian distance.

$$ED_i = \sqrt{\sum_{j=1}^{D}(x_{i,j}^{g} - x_{best,j}^{g})^2} \tag{11}$$

The individuals will be sorted according to the ED_i and classified into two niches. The first niche includes the best individuals, and the rest are in the second niche with an equal size. The population size in the gth generation is reduced by (A102). Meanwhile, a restart mechanism will be invoked when the population size is reduced to a predefined number. A new population will be generated by reinitializing the worst 10 individuals. In [6], a new adaptive population sizing mechanism is proposed, which adaptively reduces the population $pmax$ times. The jDEbin strategy is employed in the first reduction of the population and the jDEexp strategy is utilized in the second reduction. In each reduction process, the two divided populations compete with each other, and the survivor proceeds with the reduction in the next generation. If there are NP individuals in the iteration, the reduction number will be NP-1.

In order to balance the E-E of the algorithm, Choi *et al.* [84] introduced a population size reduction (PSR) mechanism to enhance the performance of ACDE [44], in which a large population size was implemented to strengthen exploration, and a small size was employed to intensify the exploitation. The population size was reduced by half in each generation [49].

Besides the linear reduction formulas above, some researchers have also proposed non-linear reduction mechanisms to adjust the NP dynamically. In order to slow down the re-

duction speed of NP at the beginning of the evolution in L-SHADE [4], Meng *et al.* [101] proposed a parabolic reduction mechanism to adaptively reduce the population size, see (A103). Similarly, Stanovov *et al.* [61] proposed a non-linear population size reduction (NLPSR) to update NP by (A104). The experimental results indicated that the NLPSR benefited the algorithm for efficient evolution. A non-linear reduction strategy was proposed by Zhang *et al.* [102] to dynamically adjust the population size, which implemented a hyperbolic tangent function by (A110). The population reduction mechanism combined with mutation adaptation and parameter adjustments helped the algorithm enhance the convergence performance.

5.2. Population augmentation

The performance of an algorithm can also be improved by increasing the population size. In [103], a self-adaptive population sizing DE with relative encoding (DESAP-Rel) mechanism was proposed, in which the population was initially set to $10{\cdot}D$, then updated by (A107). Later, Teng *et al.* [39] improved the adaptive population sizing mechanism by implementing a growth rate, see (A108). Extensive experimental results revealed that the DE variant with self-adaptive population size mechanism and relative encoding (DE-Rel) outperformed DE algorithm with a fixed population.

A new population size adaptation strategy was proposed in [104], in which the population size NP was adjusted according to the improvement obtained within L successive generations. If the best individual $\mathbf{x}_{best}$ was not improved for L generations, a new individual $\mathbf{x}_{new}$ would be generated by the mutant "DE/rand/1". If NP was less than $maxNP$, $\mathbf{x}_{new}$ would be added to the population. Otherwise, the better individual from $\mathbf{x}_{new}$ and the current worse individual $\mathbf{x}_{w}$ would be maintained. If the $\mathbf{x}_{best}$ improved more than once within L generations, and if the population size was larger than the minimum population $minNP$, the NP would be reduced by one. A similar mechanism was proposed in [105], in which the newly generated individuals would be added to the population if the best individuals were not improved. However, if the best individuals did not improve for certain generations, the newly added individuals would be removed.

Different from Ref. [104], a new population sizing strategy was proposed in [8] that the population size would be increased by one each time to enhance exploration when the algorithm obtained a better individual. Otherwise, the population size would be reduced to enhance the exploitation by removing the worst individuals. If the population was not improved for generations, the population size would be increased.

An elite-based population-incremental strategy (EPIS) was utilized in [99] to perturb the population and generate new individuals. By adopting EPIS, the elite individuals will be randomly selected and preserved in the elite archive to enhance the perturbation beyond the neighborhood. If there is no improvement for several generations, new individuals

will be generated from the top $NP{\cdot}100p\%$ best individuals and added to the population. Meanwhile, if NP reaches the lower bound (LB) for consecutive K_2 generations, the population-incremental strategy will be triggered. The experimental results indicated that JADE-APTS outperformed several newly developed DE variants.

Inspired by the natural law of survival of species that when food resources are abundant the number of species will increase, whereas when food resources are insufficient the population will shrink, Polakova [106] proposed an adaptive population sizing mechanism based on population diversity as follows.

$$DI = \sqrt{\frac{1}{NP}\sum_{i=1}^{NP}\sum_{j=1}^{D}(x_{ik} - \bar{x_k})^2} \tag{12}$$

where $\bar{x_k}$ is the mean of the kth individual in the current generation. The population size will be changed when the current relative diversity (RD) is less than $0.9rRD$ or larger than $1.1rRD$ (r is a constant number). Accordingly, the population will be increased or decreased by one. The empirical results indicated that the mechanism helped the algorithm perform more effectively compared with linear reduction [106].

Interestingly, Kitamura and Fukunaga [107] monotonically added better individuals to the population without deleting any individual in unbounded DE (UDE). They found that the replacement of population in UDE was not necessary to obtain good performance.

5.3. Population division

Cooperative search with multiple subpopulations is beneficial. Ramos *et al.* [11] extended JADE by utilizing a multiple population to solve high dimensional problems (M-PADE), which successfully helped prevent stagnation and premature convergence. The population in MPADE is divided into 15 subpopulations which adaptively adjust the JADE techniques to work in a multi-population manner. The best individuals in each subpopulation adaptively migrate and exchange information every MF generations.

$$x_{i,j}^g = 0.5 * y_{best,j}^g + 0.5 * z_{best,j}^g \tag{13}$$

where $y_{best,j}^g$ and $z_{best,j}^g$ are the best individuals from two randomly selected subpopulations **Y** and **Z**, respectively. The experimental results indicated the superiority of MPADE on the hybrid function optimizations.

The population in [108] is divided into four subpopulations, and four mutation schemes for each subpopulation are adopted: 1) DE/rand/3 (A1); 2) DE/best/3 (A2); 3) DE/rand-to-current/2 (A8); and 4) DE/rand-to-best and current/2 (A9). However, the empirical results indicated that ensemble of different adaptive schemes could not ensure significant improvement of the optimization performance.

Similarly, in [109], three indicator subpopulations and one reward subpopulation are employed to generate offsprings. The three indicator subpopulations employ three mutation schemes (DE/rand/1, DE/current-to-rand/1, and DE/current-to-$p$best/1) independently to compete with each other. The mutation scheme that performs the best will be implemented by the reward subpopulation which is larger than the indicator population. The experimental results indicated the proposed method obtained competitive experimental results.

The whole population in [47] is divided into two groups, in which the elites and non-elites are selected to generate new offsprings. The two strategies help the new algorithm balance global exploration and local exploitation capabilities. Wu *et al.* [110] proposed an ensemble of three DE variants (JADE, CoDE and EPSDE) to devise a new algorithm named ensemble of DE variants (EDEV), in which the population was adaptively divided into four subpopulations as in [109]. Every DE variant had a randomly allocated subpopulation, a reward one would be allocated to the DE variant which was prior to other DE variants within a run. After G_n generations, the best performing DE variant i_{best} will be selected.

$$i_{best} = argmax(\frac{\Delta f_i}{\Delta fes_i})_{i=1,2,3} \tag{14}$$

where Δf_i and Δfes_i are the accumulated fitness improvements obtained by the ith DE algorithm and the generations consumed, respectively. The empirical results indicated that the ensemble of DE algorithms was helpful for improving the computational results in solving several benchmark problems. In [111], the population is adaptively divided into several niches with different sizes at different stages based on diversity (distance) as in (A123).

To improve the exploration ability in early stage and enhance the convergence speed at later stages, Zhan *et al.* [112] proposed a master-slave three-population strategy, including superior, inferior and normal populations. The superior individuals were fixed to 20% of the population, while the proportion of inferior individuals would be updated by (A109). Meanwhile, the size of the whole population was updated by monitoring the improvement of the best solution. The whole population would be adaptively increased or reduced by NP_i individuals if the improvement percentage of the best solutions was larger than the threshold θ.

$$\theta = 10^{-1-4*g/G_{max}} \tag{15}$$

The empirical data indicated that the proposed method was superior to several similar techniques.

The population in [113] is adaptively divided into three subpopulations: stagnation population, converging population, and converged population. The three subpopulations

are evolved with different mutation schemes. If the stagnation subpopulation stops improving for several generations, an equiprobability distribution will be triggered. These strategies help the population remain diverse. Similarly, the population in [31] is divided into ns groups. The individuals in each subpopulation include three types: elite, mediocrity (medium), and inferior. Their offsprings are generated according to three breeding strategies (mutations). If the population is unable to find more promising solutions, new individuals will be added to the population. Conversely, the inferior individuals will be eliminated if the problem landscape is easy to optimize. The empirical results indicate that the two adaptations are conducive to accelerating the convergence. A similar population sizing mechanism is adopted in [55] (A115).

The population in [35] is adaptively divided into multiple niches on the basis of fitness values and the Euclidian distance of the individuals. Meanwhile, adaptive local search and adaptive population reduction operations are employed to obtain promising solutions and eliminate unfavorable individuals during the evolutionary process. The extensive empirical results indicated the superiority of the proposed algorithm.

Apart from dividing the population into equally sized subgroups, some researchers have also reported that adaptively dividing the population into different sizes is beneficial. In order to enhance the exploration ability, an adaptive estimation distribution (AED) mechanism was proposed in [29], by which each individual in the population would generate new individuals to form a niche. The niche size was adaptively adjusted according to the distance between individuals. In [114], the subpopulation (niche) size is adaptively changed according to its previous performance. If the niche performs well, the size of the niche will be enlarged, and vice versa.

Recently, Sun *et al.* [115] proposed an adaptive population division strategy (A111) and a dynamic population size reduction mechanism (A101) to diversify the population and enhance the convergence performance.

Cui *et al.* [116] divided the population into three sub-populations according to the sorted fitness values, the inferior, the medium and the superior. The three groups evaluate independently, with different mutation schemes and crossover rates. Meanwhile, a replacement of $a\% \cdot rand \cdot N_p$ worst individuals will be triggered when $rand < (g-1)/G_{max}$. To avoid stagnation, the population size in [117] is generated by two-stage adaptations. In the first 3/4 of G_{max}, the population size is updated by (A125), and in the last 1/4 stages it will be resized by (A126). Empirical results indicate the mechanisms benefit enhancing the accuracy and convergence speed on high-dimensional problems. Meanwhile, some researchers have adopted adaptive strategies to update the size of NP during the whole evolution process. Inspired by the implementation of F and CR adaptation mechanisms in JADE-based algorithms, Cao *et al.* [117] adopted the Lehmer mean to update the population size according to the historical improvement information and generation number.

The formulas of the adaptation mechanism of NP can be found in (A118). The adaptive population sizing strategies can be summarized in Table A3.

6. Search space adaptation

Search space or search range is the problem landscape within the lower and upper bounds of the problem. Recently, it was found that algorithms can be improved by adaptively changing the search space.

6.1. Space shrinking

The shrinking space mechanism was first proposed by Aguirre *et al.* in [118], in which the search space was shrank according to some feedback information and directed the search effort to promising regions. Inspired by [118], Fu *et al.* [15] introduced a shrinking space method (SSM) to reduce the redundant search region and enhance the convergence performance of DE. The main idea of SSM can be described as in (A127). By employing the shrinking space mechanism, the feasible search space around the potential solutions will be adaptively updated. The experimental results showed the robustness and high accuracy of the algorithm in dealing with constrained optimization problems.

A new version of DE called DEahcSPX was proposed by Noman and Iba [13], which adaptively determined the length of crossover-based local search (XLS) by incorporating an adaptive hill-climbing heuristic (AHC) and a simplex crossover (SPX) mechanisms. The experimental results revealed the superiority of the AHC mechanism to accelerate the convergence velocity and perform better than other XLS methods.

In order to enhance the search ability on multi-peak problems, Zhang *et al.* [119] proposed a mechanism named "sampling-based resizing search space" to adaptively narrow the boundaries. The variables will be resized by the minimum and maximum values of the top N_t individuals in ascending order, which can be presented as (A141).

To alleviate the limitation that the BetaCOBL strategy generates solutions outside the search space, an adaptive BetaCOBL (ABetaCOBL) mechanism was proposed in [120]. The boundary of the search space will be updated by x_{min} and x_{max} (A134). The A-BetaCOBL strategy enables the algorithm to adaptively narrow the search space, which is proven efficiently to focus on promising regions to find the optimal solution.

Jia *et al.* [121] proposed chaotic local search (CLS) with a shrinking search space mechanism to improve the exploitation. The CLS formula in the gth generation is described in (A132). The mechanism helps to explore large landscapes at early stages and exploit small regions at later phases.

To improve the convergence speed and prevent from falling into local optima, Li *et al.* [122] reduced the search space for every N_c consecutive generations. The lower and upper

bounds of the solution space will be updated by (A142). The empirical data showed that shrinking the search space helped the algorithm avoid falling into local optima.

Similarly, an adaptive scaling factor λ introduced in [123] was utilized to adaptively reduce the search space according to the improvement or stagnation of the algorithm within every generation. Meanwhile, a rollback mechanism would be triggered to return to the former search domain if the convergence performance was not improved for consecutive N_c generations. The experimental results verified the improvement in the convergence speed and precision of the computation.

6.2. Space extension

The boundary extension by mirroring (BEM) method was first proposed by Tsutsui [124] which allowed the individuals to be located beyond the boundaries of the search domain with a probability r_e, $r_e \in (0.2, 0.3)$. The new individuals generated by BEM are according to (A147). The experimental results indicated the improvements of the optimum, especially at the corner of the search space.

Similarly, Miao *et al.* [125] proposed an adaptive step size of local searching which can be defined as in (A148). The experimental results indicated that large SL benefited for global search at the beginning of the iteration, while small SL helped local search in later generations.

A space adaptive (SA) DE algorithm (ISADE) proposed in [126] was adopted to avoid blind search and premature convergence during the evolution process. The idea of SA includes two phases: expansion and contraction, which are demonstrated in six cases in [126]. The experimental results indicated the superiority of the algorithm on the convergence speed and computational accuracy. Zhang *et al.* [127] proposed an evolutionary scale adaptation (ESA) mechanism to adaptively narrow or expand the search scale. ESA is a simple yet effective method to adaptively select the nearest or farthest trail vector. The evolutionary scale indicator ψ is updated by (A149). Different from many cooperation strategies at the individual level, the adaptive dimension level adjustment (ADLA) mechanism proposed by Deng *et al.* [128] was at the dimension level. The mechanism would be triggered when the fitness of the population did not improve for consecutive generations. The ADLA includes two search space adjustment operations as in (A143-A146).

6.3. Space division

Inspired by the estimation of distribution algorithm (EDA) [129], Zhu and Yan [14] proposed an adaptive variable space differential evolution algorithm (AVSDE) which guided individuals to search in adaptive variable space (AVS) to improve the global optimization performance. In [14], AVS helps to extract global information from population distribution in the search space and leads the evolution. The variable space is determined

based on the statistical information of AG^d (advantage group, $d \in (0, 1]$ is a probability of the best individuals). The variable space Ω^k in [14] is divided into three subspaces: $\Omega_{j,i}^{k-1}$ (i=1, 2, ..., D; j=1, 2, 3). The distribution probability $p_{j,i}^{k-1}$ is updated by

$$p_{j,i}^{k-1} = \frac{m_{j,i}^{k-1}}{m_1} \tag{16}$$

where $m_{j,i}^{k-1}$ is the population size of AG^d, m_1=round($d \cdot NP$). The lower and upper bounds of the kth variable space are adaptively updated according to the value of $p_{j,i}^{k-1}$. The conditions which trigger the space variation are well defined in (7) in [14]. Meanwhile, an expansion mechanism is employed to extend the variable space. If $\sum_{j=1}^{3} p_{j,i}^{k-1} < \beta$ ($\beta < \alpha$), then the process will be repeated to previous steps.

To solve expensive optimization problems with discontinuous responses (DOPDRs), Wang *et al*. [130] proposed a region division mechanism to divide the solution space into subregions. The support vector machines (SVMs) are trained to determine the subregions. The region is divided based on the so-called objective-decision distance (ODD) as (A157).

Recently, Sun *et al.* [16] proposed an adaptive regeneration strategy based on search space adjustment (ARSA), which included two space adjustment strategies for population reinitialization: 1) the individual itself update (IIU) scheme; and 2) the optimal individual update (OIU) scheme. The IIU and OIU are defined as in (A153)-(A156). The empirical results indicated that the ARSA framework would significantly enhance the convergence speed and accuracy of some DE algorithms.

The search space in [131] is divided into many overlapped subspaces according to principal component analysis (PCA) and random decision variable (RDV) selection. Each subspace will be optimized by a surrogate model. The details of the two subspace construction mechanisms can be found in Algorithm 1 and Algorithm 2 in [131]. The empirical results indicated that the adaptive subspace mapping strategy helped enhance the convergence. Zhang *et al.* [132] proposed a new technology inspired by signal processing. They introduced a domain transform mechanism to transfer the solution domain to frequency domain. After deleting few highest frequencies, the remain are inversely transformed to solution domain. The adaptive search space strategies can be summarized in Table A4.

7. Adaptive learning strategy

A learning strategy adopts feedback from previous knowledge which is an effective method to improve the performance of EAs [133]. As shown in Fig.3 [134], a learning scheme can be described as follows: a learning cycle that commences with an output $\boldsymbol{\alpha}$ from the environment, the receiver receives the stimulus and adaptively adjusts the inputs with the learning scheme to output new stimulus $\boldsymbol{\beta}$ for the environment $\boldsymbol{\gamma}$. The

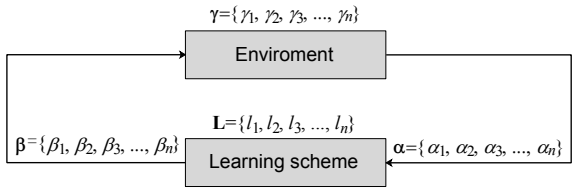

Fig. 3. Learning strategy.

environment then automatically adjusts its parameters to the stimulus $\boldsymbol{\beta}$ and generates a new $\boldsymbol{\alpha}$ for the learner. The output and input will be periodically updated until the stopping criteria are satisfied. Learning strategies can be manifold.

7.1. Historical learning

Learning from historical records benefits the algorithm from approaching the global optimum more efficiently. Since Tanabe and Fukunaga [42] proposed the SHADE algorithm, in which historical information was recorded and inherited from the parents, many more SHADE-based DE variants have been proposed, such as [4, 10, 34, 66].

A historical experience-based mechanism learning (HEML) strategy proposed in [38] helps to adaptively update the historical successful rate (SR) τ_j of the jth mutation strategy as in (A159). The historical and heuristic mechanisms help the proposed algorithm select more suitable mutation strategies and parameters at different evaluation stages.

In addition to the control parameters, some other factors can also be adaptively updated by learning strategies. In [135], a learning mechanism was proposed in which the probabilities of selecting a candidate solution generation strategy would be updated adaptively. The selection probabilities were adaptively adjusted by the accumulated success rate using a learning strategy. The selection probability of the jth strategy in the gth generation was initially set to $1/N_{cs}$ (N_{cs} is the number of candidate strategies) and then updated by

$$p_j^g = \frac{\mathbf{S}_j^g}{\sum_{j=1}^{N} \mathbf{S}_j^g} \tag{17}$$

where $\mathbf{S}_j^g$ is the matrix that stores the successful candidates generated by the jth strategy within a learning period. Otherwise, the number of discarded candidates generated by the jth strategy within N_{cs} generations will be stored in the matrix $\mathbf{M}$. A similar learning mechanism can also be found in [18].

In order to overcome mis-interaction the of the control parameters in JADE [3] and SHADE [42], Meng *et al.* [12] proposed parameters with an adaptive learning mechanism (PALM) in which the population was divided into k groups by employing the stochastic universal selection technique [136]. Each individual in the jth group employs the same F_j

and CR_j which are adaptively tuned by learning from the historical performance of the population.

7.2. Group learning

In order to improve the search performance of JADE, Takahama *et al.* [137] proposed a group-based learning (GBL) of probability density function (PDF) scheme. In GBLDE, the population is divided into K groups, and individuals learn from different groups. The group ID of an individual x_i is defined by

$$group(x_i) = \frac{r_i}{NP}K \tag{18}$$

where r_i is the sorted number of the ith individual. The empirical results indicated that the new approach was competitive with or superior to JADE.

To avoid time-consuming manual effort in robot control, Juang *et al.* [138] proposed an adaptive group-based DE algorithm (AGDE) to adaptively tune the fuzzy controller (FC). A cost function was adopted to evaluate the learning performance of the FC. Based on the performances during the iteration, the population was adaptively grouped into different clusters. Meanwhile, a group-based mutation scheme was adaptively utilized to improve the performance of the optimization.

Similarly, a preferred learning strategy proposed in [45] was used to improve the mutation operation in DE algorithm as in (A162). The differential grouping mechanism was adopted to identify the interdependence of different decision variables. The empirical results indicated that the preferred learning strategy and adaption mechanism employed in [45] helped the algorithm to enhance convergence speed and accuracy.

7.3. Opposition-based learning

Tizhoosh and Hamid [77] proposed the opposition-based learning (OBL) mechanism which searched the mirrored position. For a D-dimensional variable **x**=$(x_1, x_2, ..., x_D) \in R^D$, $x_j \in [a_j, b_j]$, the opposite point $\bar{\mathbf{x}} = (\bar{x}_1, \bar{x}_2, ..., \bar{x}_D)$ can be defined as in (A163). The opposite learning mechanism provides more chances to find the global optimum in the search space.

To improve the adaptiveness of the OBL mechanism, a generalized OBL mechanism (GOBL) was proposed by Wu *et al.* [139] in which a space transfer factor k was employed to (A163). The opposite point can be redefined as in (A164). The OBL and GOBL utilize a fixed probability of opposition (p_o) to search new landscapes. The effectiveness of OBL and GOBL is validated in [77] and [139], respectively.

In order to relieve premature convergence as the population becomes small and less diverse in the later search processes, Choi *et al.* [140] proposed an improved DE with elite

opposition-based learning (EOBL) to diversify the population and improve the optimization performance. The EOBL scheme can be described as in (A164), which implements an elite population rather than all the individuals. The experimental results indicated that the EOBL mechanism was beneficial for improving the algorithm performance.

Rahnamayan *et al.* [141] accelerated DE by utilizing the OBL and a generation jump mechanism. In [141], the opposite population P_o is generated by (A163), and the population is composed of NP fittest individuals selected from P_o. Meanwhile, a generation jump probability J_r is utilized to avoid premature convergence. Unlike in Ref. [77] and Ref. [139] which employ fixed opposition probabilities, Liu [20] proposed an adaptive mechanism to adaptively update the opposite learning probability p_o according to the success rate of the opposite learning operation. p_o is defined as in (A165). The p_o varies according to the success rate sr by adopting the opposition learning operation. If better solutions are discovered, the value of p_o will increase, and vice versa. Similarly, Song *et al*. [142] modified the OBL strategy in their algorithm by (A168), in which the base vector changed to the current best individual.

7.4. Machine learning

Recently, some researchers have attempted to adaptively tune the control parameters by machine learning. Li *et al*.[143] proposed a reinforcement learning (RL) mechanism with a fitness-ranking mechanism to guide the convergence. The RL is based on Q-learning, which adaptively updates the Q values by (A167). The experimental results indicated that DE combined with RL significantly helped improve the search efficiency. As a classic technology, Q-learning enables offline learning and helps to select optimal actions in the Markov processes. Different from the methods mentioned above, three DE strategies and four CRs are encapsulated as actions and recorded in the Q-table in [144]. The states are adaptively divided according to the population size. Each individual acts as an agent in RL.

Similarly, a deep Q-learning network (DQN) proposed in [145] was trained to adaptively select proper mutations. The Q-values were updated by (A167), a deep RL that extended Q-learning by employing a deep neural network (DNN). It was trained by analyzing the running results on several benchmarks. The weights of DNN were adjusted by minimizing a loss function.

$$minimize\ [r^{target} - Q(S_j, A_\theta)]^2 \tag{19}$$

where r^{target} is a reward value, γ is the discount factor, θ is the weight in Q-learning. The experiments indicated the competitiveness of the proposed algorithm.

In order to learn from optimization performances, Sun *et al*. [146] employed a deep neural network (DNN) to adaptively adjust F and CR. Meanwhile, an RL policy gradient mechanism was applied to optimize the parameters of DNN, which was based on

a finite-horizon Markov decision process. Empirical results indicated the competitiveness of the proposed approach based on the DNN mechanism. However, the parameter training was difficult and time-consuming. To balance the exploitation and exploration, a Q-learning methodology is suggested by Lu *et al*. [147] to adaptively combine the mutation and crossover mechanisms with an updating Q-table. Meanwhile, to diversify the population, an ensemble of three splitting population strategies are employed in the initialization process. In [148], a reinforcement learning-based adaptive resource allocation (RLRA) mechanism, is employed to allocate the FEs according to the fitness-and-distance Q-table. Empirical results show that RLRA strategy benefits multimodal optimizations.

The adaptive learning strategy can be summarized in Table A5.

8. Composite adaptations

Many adaptive DE algorithms employ multiple strategies to remain diversity, and hence prevent from premature and stagnation. In this paper, we categorize the algorithms which employ more than three classes of adaptive strategies into the composite adaptation.

8.1. Composite of control parameter and mutation strategies

Typically, control parameters and mutation adaptation schemes are the most frequently adopted in adaptive DE algorithms. The ensemble of various strategies is frequently used in mutation adaptation methodologies.

8.2. Composite of control parameter, mutation, and population strategies

In addition to mutation pooling strategies, ensembles of populations are also reported beneficial. Based on JADE, two mutation schemes, population reduction and augmentation strategies are implemented in [8]. Based on jDE, Zamuda *et al.* [9] employed multiple mutation strategies in conjunction with a structured population reduction mechanism to successfully improve the original DE variant.

In [14], proper mutation schemes are adaptively selected from an ensemble of three strategies. Meanwhile, F and CR are adaptively generated from a normal distribution, and a dynamic search space mechanism is implemented to improve the convergence speed.

Meanwhile, the linear population size reduction mechanism employed in LSHADE [4] is also implemented in [12]. The experimental results indicated the superiority of LPALM-DE on the 44 benchmark function evaluations. An ensemble of three mutations was implemented in [6] to enhance the search ability. F and CR are adaptively updated by (A22) and (A23), respectively. Meanwhile, a population size reduction mechanism is employed in the algorithm to successfully solve large-scale optimization problems. The mutation schemes are adaptively selected from an ensemble in [17], where the parameters

F and CR are adjusted using a uniform distribution combined with sigmoid functions. This process is accompanied by an OBL mechanism to diversify the population. The experimental results indicated that it was helpful to deal with real-world optimizations. Apart from the adaptation of control parameters, ensemble of four mutation candidates and different population sizes are employed in [7]. Wu *et al.* [149] comprehensively surveyed the ensemble of strategies employed in population-based algorithms, suggesting two levels of ensembles, three techniques to realize the ensembles, and ensemble methods for different problems.

Zhan *et al.* [112] developed an adaptive distributed DE (ADDE), in which F and CR were adapted by Cauchy and Gaussian distributions, respectively. Moreover, a master-slave multipopulation strategy consisting of three types of slave subpopulations, along with three mutation strategies, was implemented to enhance the search capability.

In [93], the control parameters are changed by two-stage self-adaptive strategies. Three mutation mechanisms are employed according to their successful probabilities. Meanwhile, the population size is resized by (A105). Gong *et al.* [150] employed both population division and linear reduction mechanisms to adaptively balance the search directions. Besides, adaptive multiple mutation and crossover strategies are employed.

Based on jSO, Zhou and Huang [151] made three main improvements in NLPSR-jSO: 1) The linear populations strategy is changed to non-linear strategy as in (A106); 2) An adaptive archive usage (AAU) is proposed to prevent from overusing the archive; 3) The rank-based selective pressure strategy (RSP) used in LSHADE-RSP [152] is modified to avoid overusing of dominant individuals. The control parameters F and CR are updated as in jSO [68]. Empirical results indicate its significant improvements in computational accuracy and convergence speed.

8.3. Composite of control parameter, mutation, and learning strategies

Rather than random selection of a mutation scheme from a candidate pool, Tan and Li [145] proposed a deep Q-learning mechanism to train a Q-learning neural network, which was able to adaptively select a proper mutation strategy according to the problem features.

PALM-DE [12] simultaneously utilizes a parameter adaptive learning mechanism (PALM) to adaptively adjust the controlling parameters F and CR, and the mutation schemes employed in JADE [3].

8.4. Other ensemble of adaptive strategies

Apart from implementing control parameter and mutation adjustments, Liu *et al*. [38] employed historical learning and heuristic information for adaptations in HHDE. The population in HHDE was evaluated by adaptively adjusting the control parameters and three mutation strategies, associating with historical and heuristic information.

Similarly, an ensemble of five adaptive strategies was proposed in the local binary pattern-based adaptive DE algorithm [70]. The adaptive strategies implemented in [70] included: 1) fitness-based F adaptation; 2) distance-based CR adaptation; 3) neighborhood mutation; 4) adaptive population division; and 5) historical learning. The experimental results indicated that the proposed technology was superior or competitive on multimodal optimizations. Nishihara and Nakata [153] developed a hybrid DE algorithm that integrated multiple strategies, including: 1) randomly selecting from the four DE strategies (A2), (A3), (A5), (A66); 2) different CR strategies; 3) population division; 4) prior validation based on fitness improvement; and 5) post hoc validation strategies to generate effective parameters. The empirical results demonstrated that the ensemble of these mechanisms enabled the proposed algorithm to effectively tackle computationally expensive problems.

Recently, Song *et al.* [142] proposed an ensemble of multiple strategies in OLBADE, in which F and CR were adaptively generated by (A27) and (A28), respectively. The donor vector was dynamically perturbed during the evolution. The population size was adaptively adjusted by (A102). Meanwhile, the OBL strategy was adopted to diversify the population. The empirical results indicated that the multiple strategies benefited the algorithm in its effectiveness and robustness. The composite adaptations can be summarized in Table A6.

There are six main strategies proposed in [150], including: 1) ensemble of three DE strategies; 2) adaptive crossover operator; 3) population division; 4) linear population size reduction; 5) parameter adaptation strategies; and 6) superior-trial-involved survivor mechanism. Empirical results indicate that these strategies help the algorithm perform well on high-dimensional hybrid and composition problems.

9. Comparison

Based on an extensive survey, the typical strategies employed in adaptive DE algorithms and their complexities can be summarized in Tables 2 and 3. The algorithms are categorized according to the strategies mainly adopted. If an algorithm incorporates more than three categories of strategies, it will be classified as a composite adaptation.

Based on comparisons and analyses, several intriguing findings can be summed up as follows:

1) Most adaptive DE algorithms employ multiple strategies, frequently incorporating the adaptation of control parameters and mutation techniques.

2) Cauchy and Gaussian distributions are the most commonly used methods for adaptively adjusting F and CR, accounting for 26% and 40% of the selected algorithms in Tables 2 and 3, respectively.

Table 2
Comparison of typical adaptive strategies in DE variants

Adaptation	Algorithm	F	CR	Mutation	Population	Space	Learning	Other mechanism	Complexity
Parameter	MSDE [18]	Normal	Normal	DE/rand/1/bin	fixed	fixed	historical	accumulate F in L runs	**O**(Gmax·NP·D)
Parameter	ACDE [44]	Cauchy	Cauchy	DE/rand/1/bin	fixed	fixed	historical	F and CR memory	**O**(Gmax·NP·D)
Parameter	aDE [54]	Uniform	Uniform	DE/rand/1/bin	fixed	fixed	none	fitness-based parameters	**O**(Gmax·NP·D)
Parameter	AEPD-JADE [154]	Cauchy	Normal	DE/current-to-pbest/1/bin	fixed	fixed	none	population diversity	**O**(Gmax·NP·D)
Parameter	GADE [69]	greedy scheme	greedy scheme	DE/rand/1/bin	fixed	fixed	neighborhood learning	periodically update the parameters	**O**(Gmax·NP·D)
Parameter	JADE [3]	Cauchy	Normal	DE/current-to-pbest/1/bin	fixed	fixed	none	external archive	**O**(Gmax·NP·[D+log(NP)])
Parameter	jDE [2]	Uniform	Uniform	DE/rand/1/bin	fixed	fixed	none	parameter for each individual	**O**(Gmax·NP·[D+log(NP)])
Parameter	p-ADE [155]	Cauchy	Cauchy	DE/rand-to-best/$p$best/bin	fixed	fixed	none	3 adaptive scale factors	**O**(Gmax·NP·D)
Parameter	FiADE [32]	fitness-based	fitness-based	DE/best/1/bin	fixed	fixed	none	none	**O**(Gmax·NP·D)
Parameter	DEIDM [56]	fitness-based	fitness-based	new DE mutation	fixed	fixed	none	individual-dependent mechanism	**O**(Gmax·NP·D)
Parameter	PVADE [156]	distance-based	distance-based	DE/rand/1/bin	fixed	fixed	none	quasi-opposition	**O**(Gmax·NP·D)
Parameter	SDE[30]	parameter ensemble	parameter ensemble	mutation ensemble	fixed	fixed	none	probability for parameter selection	**O**(Gmax·NP·D)
Parameter	FADE [63]	fuzzy logic	fuzzy logic	DE/rand/1/bin	fixed	fixed	none	fuzzification and defuzzification	**O**(Gmax·NP·D)
Mutation	SaDE [5]	Normal	Normal	mutation ensemble	fixed	fixed	historical	median value for Normal	**O**(Gmax·NP·D)
Mutation	EFADE [157]	Uniform	Uniform	Triangular mutation	fixed	fixed	historical	new scaling factor adaptation	**O**(Gmax·NP·D)
Mutation	CoDE [37]	parameter ensemble	parameter ensemble	mutation ensemble	fixed	fixed	none	fixed pair of parameter settings	**O**(Gmax·NP·D)
Mutation	EPSDE [158]	linear	linear	mutation ensemble	fixed	fixed	none	randomly select mutation scheme	**O**(Gmax·NP·[D+log(NP)])
Mutation	SaJADE [33]	Normal and Cauchy	Normal	mutation ensemble	fixed	fixed	none	inferior archive	**O**(Gmax·NP·D)
Mutation	ISADE [19]	sigmoid	Uniform	mutation ensemble	fixed	fixed	none	adaptively select mutation scheme	**O**(Gmax·NP·D)
Mutation	SMADE [159]	Cauchy	Normal	mutation ensemble	fixed	fixed	none	variance-based adaptation	**O**(Gmax·NP·D)
Mutation	DEPSO [89]	Uniform	Uniform	mutation ensemble	fixed	fixed	none	elite archive	**O**(Gmax·NP·D)
Mutation	LSHADE-Code [93]	Cauchy	Gaussian	mutation ensemble	adaptive reduction	fixed	none	dual-stage parameter adaptation	**O**(Gmax·NP·D)
Mutation	AGDE[74]	Uniform	Uniform	new DE mutation	fixed	fixed	historical	adaptive CR sets	**O**(Gmax·NP·D)
Mutation	MDE$_p$BX [76]	Cauchy	Normal	DE/current-to-gr_best/1	fixed	fixed	none	adaptive p-best crossover	**O**(Gmax·NP·D)
Mutation	SDE[30]	parameter ensemble	parameter ensemble	mutation ensemble	fixed	fixed	none	probability for parameter selection	**O**(Gmax·NP·D)
Population	APTS-DE [99]	fixed	fixed	neighborhood search	increase and decrease	fixed	historical	inferior and elite archive	**O**(Gmax·NP·D)
Population	DE-Rel [39]	Uniform	fixed	DE/rand/1/bin	increase and decrease	fixed	none	population size requires more than 3	**O**(Gmax·NP·D)
Population	DESAP [103]	fixed	uniform distribution	DE/rand/1/bin	increase and reset	fixed	none	elitism and culling is achieved	**O**(Gmax·NP·D)
Population	DEVP [104]	fixed	fixed	DE/best/1/bin	increase and decrease	fixed	none	remain the best individual	**O**(Gmax·NP·D)
Population	dynNP-DE[49]	Uniform	Uniform	DE/rand/1/bin	reduction	fixed	none	subpopulation competition	**O**(Gmax·NP·D)
Population	EsDEr-NR[100]	Sinusoidal + Cauchy	Normal	DE/current-to-pbest/1/bin	reduction	fixed	none	restart mechanism	**O**(Gmax·NP·D)
Population	IMSaDE [47]	Uniform	Uniform	DE/rand/2	division	fixed	none	better and worse subpopulation	**O**(Gmax·NP·D)
Population	jDEdynNP-F [48]	Uniform	Uniform	DE/rand/1/bin	reduction	fixed	none	dynamic sign of F	**O**(Gmax·NP·D)
Population	AED-DDE [29]	fixed	fixed	DE/rand/1/bin	master-slave multiniche	fixed	neighborhood	local search	**O**(Gmax·NP·D)
Population	DDE-ARA [160]	fixed	fixed	mutation ensemble	multi-machine	fixed	none	Flag to judge the state	**O**(Gmax·NP^2·D)
Population	DADE [111]	fixed	fixed	diversity-based mutation	diversity-based division	fixed	none	tabu archive-based local process	**O**(Gmax·NP^2·D)
Population	APSDE [36]	Cauchy	Normal	mutation ensemble	linearly decrease	fixed	historical	dual trial vectors	**O**(Gmax·NP·D)
Population	EJADE [117]	Cauchy	Normal	DE/current-to-$p$best/1	Gaussian mutation	fixed	none	population information similarity	**O**(Gmax·NP^2·D)

3) In contrast, search space is not widely employed as an adaptation mechanism. Only 8% of the algorithms listed in Tables 2 and 3 utilize space adaptation strategies.

4) Most adaptive strategies do not significantly increase the complexity of the algo-

Table 3
Comparison of typical adaptive strategies in DE variants (continued)

Adaptation	Algorithm	F	CR	Mutation	Population	Space	Learning	Other mechanism	Complexity
Space	ARSA [16]	fixed	fixed	DE/rand/1/bin	fixed	variable	none	record individual status and radius	**O**(Gmax·NP·D)
Space	ATMDE [15]	fixed	fixed	mutation ensemble	fixed	shrink	none	adaptive trade-off model	**O**(Gmax·NP·D)
Space	AVSDE [14]	Normal	Normal	mutation ensemble	fixed	variable	none	statistic analysis	**O**(Gmax·NP^2·D)
Space	DEahcSPX [13]	fixed	fixed	DE/rand/1/bin	fixed	Lamarckian local search	none	hill-climbing local search	**O**(Gmax·NP·D)
Space	DECLS [121]	Uniform	Uniform	DE/rand/2/bin	fixed	chaotic search	none	chaotic iteration	**O**(Gmax·NP·D)
Space	IADE [122]	Uniform	exponential	DE/rand/2/bin	double population	division and reduction	none	population sorting	**O**(Gmax·NP·D)
Space	ESADE [127]	Uniform	fixed	DE/rand/1	fixed	fixed	none	successful scale estimation	**O**(Gmax· (NP· D + NP· log_2 NP)
Learning	SHADE [42]	Cauchy	Normal	new DE mutation	fixed	fixed	historical	external archive	**O**(Gmax·NP·D)
Learning	AGODE [20]	fixed	fixed	DE/rand/1/exp	fixed	fixed	opposite learning	adaptive opposition rate	**O**(Gmax·NP·D)
Learning	EOBL-DE [17]	sigmoid	Uniform	mutation ensemble	fixed	fixed	opposite learning	jumping and elite OBL	**O**(Gmax·NP·D)
Learning	GBLDE [137]	Cauchy	Normal	DE/current-to-pbest/1/bin	division	fixed	group learning	based on JADE	**O**(Gmax·NP·D)
Learning	GOBL-ACDE [139]	Normal	Normal	mutation ensemble	fixed	fixed	opposite learning	cosine selection model	**O**(Gmax·NP·D)
Learning	LA-DE [134]	parameter ensemble	fixed	DE/current-to-best/1/bin	fixed	fixed	learning automata	reward-penalty scheme	**O**(Gmax·NP·D)
Learning	OBDE [141]	fixed	fixed	DE/rand/1/bin	fixed	fixed	opposite jumping	generation jumping	**O**(Gmax·NP·D)
Learning	SLDDE [135]	fixed	Normal	mutation ensemble	fixed	fixed	historical	CR memory	**O**(Gmax·NP·D·Q)
Learning	DE-RLFR [143]	fixed	fixed	mutation ensemble	fixed	fixed	Q-learning	hierarchical state	**O**(Gmax·NP·D)
Learning	DEDQN[145]	Cauchy	Normal	mutation ensemble	fixed	fixed	deep Q-learning	random walk	**O**(Gmax·NP·[D+ log(NP)]·steps)
Learning	MSDE[147]	fixed	fixed	DE/rand/1	fixed	fixed	Q-learning	ensemble population initialization	**O**(Gmax·NP·D)
Learning	LADE [146]	nonlinear	nonlinear	DE/current-to-pbest/1/bin	fixed	fixed	deep neural network	policy gradient	**O**(Gmax·NP·Q·M ·L·T·D)
Composite	jDElsgo2010 [67]	self-adaptive	uniform	mutation ensemble	reduction	fixed	none	sign of F	**O**(Gmax·NP·D)
Composite	jDEscop2011 [6]	self-adaptive	self-adaptive	mutation ensemble	reduction	fixed	none	three DE mutations	**O**(Gmax· log(NP)·D)
Composite	jDEsps2012 [7]	self-adaptive	self-adaptive	mutation ensemble	increase and decrease	fixed	none	multiple population sets	**O**(Gmax·NP·D)
Composite	LDECCDG [45]	Normal	Normal	DE/rand/1/bin	division	fixed	preferred learning	linearly reduce F and CR	**O**(Gmax·NP·D)
Composite	LSHADE [4]	Cauchy	Normal	DE/current-to-pbest/1/bin	linear reduction	fixed	none	external archive	**O**(Gmax·NP·[D+ log(NP)])
Composite	LSHADE_cnEpSin [10]	sinusoidal	Normal	DE/current-to-$p$best/1/bin	linearly decreasing	fixed	covariance matrix learning	Euclidean neighborhood	**O**(Gmax·NP·D)
Composite	MPADE [11]	Normal	Normal	varied DE/Current-to-$p$best/1	division	fixed	archive	archive to store optimal individuals	**O**(Gmax·NP·D)
Composite	PALMDE [12]	Cauchy	Normal	DE/target-to-$p$best/1/bin	linearly decreasing	fixed	historical	store best and worst individuals	**O**(Gmax·NP·D)
Composite	SapsDE [8]	Cauchy	Normal	mutation ensemble	increase and decrease	fixed	none	population resizing	**O**(Gmax·NP·D)
Composite	SPSRDEMMS [9]	Uniform	Uniform	mutation ensemble	half reduced and structured	fixed	none	an extension of jDE	**O**(Gmax·log(NP)·D)
Composite	EADE [161]	Normal	parameter ensemble	varied DE/rand/2/bin	fixed	fixed	historical	use flag to sign the improvement	**O**(Gmax·NP·D)
Composite	ANDE[113]	Cauchy	Normal	DE/best/1/bin	exclusion and augmentation	fixed	historical	heuristic clustering	**O**(Gmax·NP^2·D)
Composite	HHDE[38]	Gaussian	Gaussian	mutation ensemble	fixed	fixed	historical	heuristic information-based mechanism	**O**(Gmax·NP·D)
Composite	ADDE[112]	Cauchy	Normal	mutation ensemble	master-slave multipopulation	fixed	none	threshold based improvement for population adaptation	**O**(Gmax·N·(D+ log(N)))
Composite	LBPADE[70]	fitness-based	distance-based	mutation ensemble	division	fixed	historical	boundary constraint	**O**(Gmax·NP·D)
Composite	OLBADE[142]	Cauchy	Normal	vector perturbation	linear reduction	fixed	opposite learning	boundary constraint	**O**(Gmax·NP·D)
Composite	TPADE[150]	Cauchy	ensemble crossovers	ensemble mutations	linear reduction	fixed	none	Superior-trial-involved survivor selection	**O**(Gmax·NP·D)

rithms. However, the DE variant that employs machine learning techniques such as deep Q-learning [143] requires more computational resources and longer CPU runtime.

5) The majority of composite adaptation algorithms adopt Cauchy and Gaussian distri-

butions, along with an ensemble of mutations and adaptations in population size, to update control parameters, mutation strategies, and population sizes, respectively.

Based on empirical results from the literature, the advantages and disadvantages of the adaptive strategies in DE variants can be summarized in Table 4.

Table 4
Typical merits and demerits of different adaptive strategies.

Adaptations	Merits	Demerits
Control parameter	1) accelerate the convergence speed [3, 30, 69]; 2) enhance the performance on high dimensional problems [63, 69]; 3) help to maintain better solutions [32, 69]; and 4) enhance the E-E [156, 158].	1) the greedy parameter adaptation may lead to stagnation [2, 69]; 2) the adaptation of the parameters requires more iterations and computation time [155]; and 3) random selection of parameters may lead to local optimum [158].
Mutation	1) fast convergence [1, 33, 157, 162]; 2) enrich the diversity of the population [100]; and 3) enhance the global and local search [157].	1) deteriorate the population diversity when solving high dimensional and complex problems [2]; 2) lead to stagnation [4, 25]; and 3) greedy mutations may lead to fall into local optimum [33, 155].
Population size	1) significantly benefit for high-dimensional problem optimization [48, 99, 104, 117, 163]; 2) speed up the convergence [49, 99, 160]; 3) reduce the computation cost by continuously decreasing the population size [4]; and 4) relieve the premature and stagnation [100, 154].	1) increase the computational cost when the population is large [111]; 2) small population size hampers the performance of the algorithm on hybrid problems [163]; and 3) do not perform better on low-dimensional problems [104].
Search space	1) speed up the convergence in the later process of the evolution [13, 16, 121, 122]; 2) do not increase the time consumption [13, 14, 16]; 3) obtain superior performance on low dimensional problems [14, 16]; and 4) help to balance the E-E [16].	1) can not significantly improve the performance on high dimensional problems [14, 16]; and 2) require additional parameters which increases the computation complexity [14, 16].
Learning strategy	1) enhance the diversity of the population [134, 137, 143]; 2) benefit for solving complex problems [17, 33, 134, 162]; 3) help to achieve better solutions [4, 17, 20, 135, 143]; 4) generate better parameters [143, 145, 146]; and 5) improve the convergence speed [17, 137, 143].	1) increase the training time and computation resource[146]; 2) increase the computation complexity [145, 146]; and 3) hard to make the training stable[20, 146].
Composite adaptations	1) integrate the merits of different strategies and avoid the influence of weak strategies [110, 164]; 2) speed up convergence [70, 101, 112]; 3) enhance the diversity of the population [38, 70, 110, 112]; 4) show excellent performance when dealing with large-scale and hybrid problems [4, 6, 7, 11, 45, 67, 70, 100, 110, 112, 161]; and 5) well balance the E-E [100].	1) ensemble of best techniques can not ensure the superiority [7, 8, 110, 112]; 2) increase the complexity of the algorithm [9, 26, 112]; and 3) difficult to select and integrate proper adaptations.

As illustrated in Table 4, all six categories of adaptive strategies are reported to be beneficial for accelerating convergence. Control parameter and population size adaptations help enhance high-dimensional problem optimizations. Both mutation and learning adaptations are noted to be helpful in enhancing the diversity of the population. Adaptive mutation, learning strategies and composite adaptations are reported to enhance the diversity of the population. However, adaptations in control parameter and mutation strategies may also lead to stagnation. Search space, learning scheme and composite adaptations

can increase the complexity of the algorithm. The comparisons indicate that adaptive strategies do not perform consistently across different problems. Therefore, they should be carefully selected and implemented.

10. Discussion

10.1. Challenges and promising research directions

10.1.1. On control parameter adaptation

Control parameter adjustment is the most frequently employed in adaptive DE algorithms. The challenge is that the best control parameter settings of DE are problem-dependent[2]. In this case, the parameters should be adaptively changed according to the problem's characteristics. It has been validated that the performance of an algorithm can be markedly improved by adaptively changing the parameters, especially on high dimensional problems [30, 63, 69]. However, the generation or selection of parameters according to a greedy strategy e.g., according to the best fitness should be avoided. Because it will lead to biased exploration and stagnation [2, 69]. Promising directions can be as follows: 1) as the value can be dynamically changed to diversify the parameters, periodic distributions such as Levy, sine, cosine, tangent and etc can be used to update F and CR; 2) the parameters can be generated or selected according to some learning strategies which are associated with the fitness value [144], the distance and the problem landscapes [71]; and 3) the BO and MAB strategies can be used to help with parameter regulation [64, 65, 95].

10.1.2. On mutation adaptation

The mutation adjustment is closely related to the control parameters, which directly helps generate new individuals. Proper strategies such as (A1)-(A5) help to diversify the population and enhance the convergence [1, 33, 162, 157]. The challenge is that the matching of parameters and mutation strategies is mainly according to the designers' cognitions. A promising direction is to generate or select appropriate mutants based on the feedback information during the evolution process, e.g. a DE variant with autonomous selection of mutation strategies and control parameters (ASDE) was proposed by Wang *et al.* [165]. The accumulated historical experience of the combination of control parameters and mutations is preserved in ASDE, which can be adaptively mapped into the rules repository and selected by the individuals. The probability matching (PM) strategy proposed by Gong *et al.* [40] was successfully used to select the most suitable mutation schemes from an ensemble. The PM strategy was based on the credit assignments on each mutation strategy according to the fitness improvements. Recently, Sun *et al.* [146] employed machine learning mechanism to successfully select the appropriate mutant, which indicated another effective method to select a suitable mutation scheme.

10.1.3. On population size adaptation

Theoretically, a large population increases the chances of finding the global optimum but suffers from huge computational cost, while a small population benefits from converging fast but confronting premature convergence [6, 103, 156]. The challenge lies in adjusting the population size to align with the specific characteristics of the problem. A promising direction is to update the population size according to a mechanism with dynamic feedback information, e.g, an annealing schedule proposed by Das and Suganthan [21] which controlled the gradually increased probability of selecting a top-ranked individual could help achieve excellent exploration at the beginning and good exploitation at the later stages. Piotrowski [163] suggested that an adaptive population sizing mechanism was highly recommended for high-dimensional problem optimizations. The adaptive division and combination of populations, along with various mutation strategies for each sub-population in [31, 110, 114] suggested another effective method to achieve better performance. Meanwhile, since feedback information can assist the algorithm in balancing the population size and performance [106], research on the variation of population size based on adaptive learning strategies warrants further investigation.

10.1.4. On search space adaptation

Search space adjustment is easier to be ignored compared with parameter, mutation scheme and population size. The successful applications of the space adjustment in [13, 16, 121, 122] indicate its effectiveness. The challenge is how to gradually narrow the search space, and hence guide the search toward smaller region that contains the global optimal solution, and finally locates the global optimum. The coefficient factors used to adjust the search domain are vital for space adaptation. Promising directions can be that the factors for adaptively controlling the search space effectively learn from the EDA [154], distribution distance [29, 100], and directional information [14].

10.1.5. On learning scheme adaptation

Learning schemes are frequently utilized to generalize or select effective parameters in DE variants. By adaptively learning from the feedback information, the parameters can be adjusted appropriately to facilitate effective searches. However, the challenge is how to implement a proper learning strategy, e.g., historical learning [3, 38, 135], group learning [45, 137, 138], opposition learning[77, 139, 140], machine learning [143, 145, 146] to learn from the feedback information during the evolution process. Meanwhile, as machine learning strategies require training processes to generate learning factors, more computational resources and longer runtime are required. A promising direction is that a learning scheme should adapt to the problems' characteristics. For large-scale and complex problems, a machine learning strategy is recommended. Otherwise, other adaptive learning strategies may be suitable.

10.1.6. On composite adaptations

As seen in Tables 2 and 3, most adaptive DE algorithms incorporate more than three strategies. It is important to investigate the correlation among various adaptive strategies. Several critical issues warrant emphasis. First, since additional mechanisms beyond adaptation strategies are adopted in most algorithms, it is challenging to identify which adaptive strategy primarily works. Second, conflicts may arise if the selected strategies are not properly employed. For instance, adopting a large population and small search space simultaneously should be avoided, and vice versa.

Third, ensembles of different adaptive strategies have been validated to improve the algorithm, particularly for large-scale problems [4, 7, 11, 45, 70, 161]. The challenge is how to adjust the search directions during the evolution process to effectively employ robust strategies for solving complex problems. One promising direction was proposed by Wu *et al.*[166] that the weights of the voters were adaptively updated according to the historical voting performance. Another promising direction is integrating adaptive strategies with distinct characteristics. The combination of different strategies can be manifold. Suppose that there are six strategies (F, CR, mutation, population size, search space, and learning scheme). According to the combination theory [167], the number of composite strategies is as follows.

$$C_6^1 + C_6^2 + C_6^3 + C_6^4 + C_6^5 + C_6^6 = 63 \tag{20}$$

Therefore, there can be 63 ensembles of different dynamic schemes! It provides evidence that more new adaptive DE algorithms can be proposed in the near future.

10.2. General framework of an adaptive (DE) algorithm

According to the no-free-lunch theory [168], different problems require various strategies to adaptively adjust the problems' characteristics. As mentioned in Eq. 20, there are many combinations of the categorized adaptive strategies. The integration of diverse adaptation strategies employed in DE algorithms can be generalized as shown in Fig. 4.

As illustrated in Fig. 4, the updating process of a DE algorithm, there can be an adaptive updating engine that includes different adaptive strategies for flexible combinations. This framework graphically summarizes how to develop or enhance a DE algorithm by implementing adaptive strategies. The framework can also be applied to other evolutionary approaches, i.e., the probabilities of crossover (p_c) and mutation (p_m), as well as the population size in genetic algorithm (GA) [169, 170] can be adaptively updated by the adaptation strategies mentioned above. The inertia weight (w) and population size in Particle Swarm Optimization (PSO) [171, 172] can be updated through the adaptive strategies employed in DE algorithms.

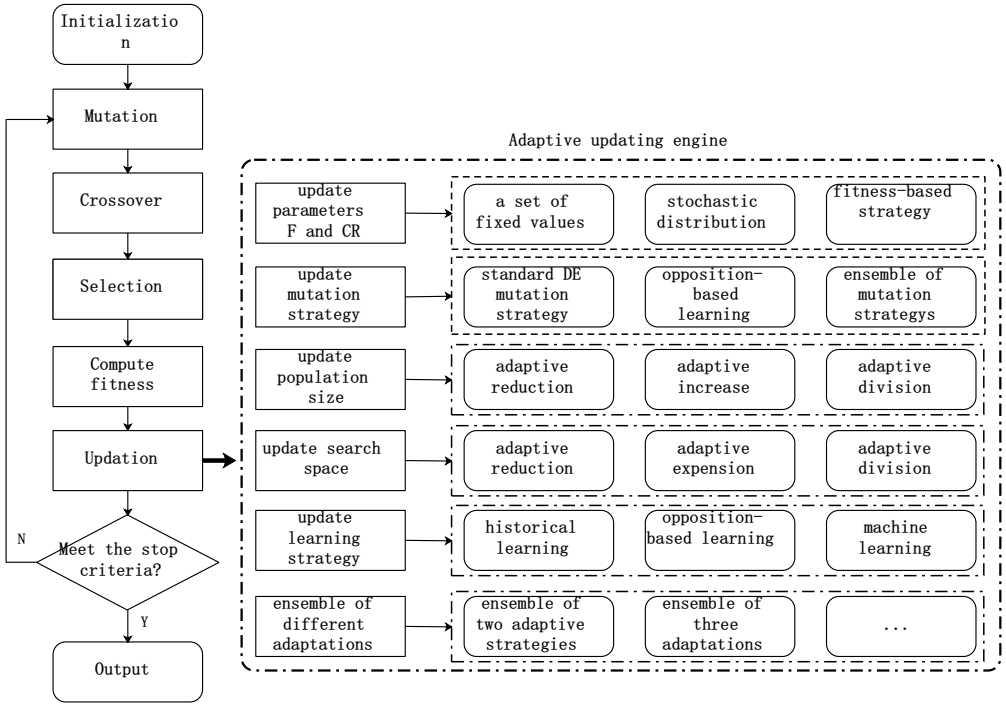

Fig. 4. A general framework of an adaptive (DE) algorithm.

10.3. Real-world applications

The application of adaptive strategies to address real-world problems is promising. As illustrated in Fig.5, the number of publications of adaptation strategies on engineering, instruments instrumentation and telecommunication problems has dramatically increased over the past five years. The data are collected from the Web of Science, where X-axes denotes subjects, Y-axes is the number of publications.

Unfolding the publications in the engineering area in Fig. 5, authors can easily find several real-world applications not represented in the figure. These include data mining [161], economic problem optimizations [109, 144], spacecraft trajectory optimization [76], manufacturing [52, 85], communication issues [36, 173], robot precisely control [174, 175], cooperative jamming weapon-target assignment (CJWTA) [135], multi-objective problem optimizations [87, 176, 143] and etc. These observations highlight the significant potential to adopt adaptive strategies to solve a broader range of real-world problems.

11. Conclusion

Adaptive strategies have been broadly implemented to successfully improve DE algorithms. To some extent, they have contributed to the prosperity of DE, which should be

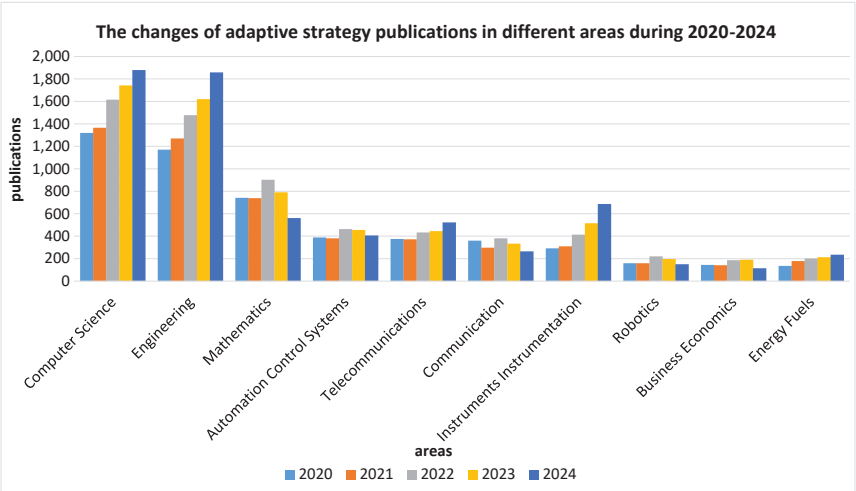

Fig. 5. The changes of adaptive strategy publications in different areas during 2020-2024.

paid more attention to unfold the mystery. This work provides a comprehensive investigations to the adaptive strategies adopted in DE variants, refining the primary adaptive operations, formulas and their impacts on the algorithms. To facilitate the identification and application of adaptive strategies, a new taxonomy that comprises six categories of adaptation strategies is proposed. A comparison of typical adaptive DE algorithms reveals that most of them incorporate multiple strategies without significantly increasing computational costs. The advantages and disadvantages of various adaptive strategies are discussed, outlining their distinct characteristics. Challenges and open issues related to these adaptive strategies are highlighted, providing promising directions for future research. The work investigates the potential for developing new adaptive (DE) algorithms by implementing diverse adaptation strategies. A general framework featuring an adaptive updating engine is proposed to assist in the design or enhancement of adaptive (DE) algorithms.

The study also reveals that the adaptive strategies mentioned above can be applied to select or fine-tune the parameters, mutation strategy, population size and search space in other meta-heuristic methodologies. Furthermore, this work also highlights the significant potential for implementing adaptive strategies to solve real-world problems that require adaptive responses. For instance, self-driving vehicles should adaptively assess the environment [117], autonomous robots need to effectively navigate around obstacles [174], vehicle-to-infrastructure cooperation requires vehicles to work adaptively in harmony with infrastructure [177], and etc. Fig. 5 unfolds the fact that the application of adaptive strategies will continue to be vibrant in multi-disciplinary research and application. Despite extensive research effort into adaptation strategies, there are still many promising jobs worth further researching, which are outlined in Section 10. Hoping that this work can be

helpful to inspire for devising more powerful adaptive algorithms.

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CRediT authorship contribution statement

Xinggui Ye: Conceptualization, Methodology, Software, Validation, Investigation, Writing -Original Draft, Writing - Review & Editing. **Jianping Li**: Supervision, Methodology, Project administration, Funding acquisition. **Peng Wang**: Conceptualization, Methodology, Validation. **Ponnuthurai Nagaratnam Suganthan**: Methodology, Investigation, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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